

Arjuna JEE 2025

Mathematics

Binomial Theorem

DPP: 5

Q1 If $(1+x)^n = C_0 + C_1x + C_2x^2 + C_3x^3, \dots + C_nx^n$,
prove that

$$C_0 + 2C_1 + 3C_2 + \dots + (n+1)C_n = (n+2)2^{n-1}$$

Q2 If $(1+x)^n = C_0 + C_1x + C_2x^2 + C_3x^3, \dots + C_nx^n$,
prove that

$$C_0 + 3C_1 + 5C_2 + \dots + (2n+1)C_n = (n+1)2^n$$

Q3 If $(1+x)^n = C_0 + C_1x + C_2x^2 + C_3x^3, \dots + C_nx^n$,
prove that

$$C_0 + \frac{C_1}{2} + \frac{C_2}{3} + \dots + \frac{C_n}{n+1} = \frac{2^{n+1} - 1}{n+1}$$

Q4 If $(1+x)^n = C_0 + C_1x + C_2x^2 + C_3x^3, \dots + C_nx^n$,
prove that

$$5C_1 + 9C_2 + 13C_3 + \dots + (4n+1)C_n = (2n+1)2^n - 1$$

Q5 If $(1+x)^n = C_0 + C_1x + C_2x^2 + C_3x^3, \dots + C_nx^n$,
prove that

$$\frac{C_0}{2} + \frac{C_1}{3} + \frac{C_2}{4} + \frac{C_3}{5} + \dots + \frac{C_n}{n+2} = \frac{1+n \cdot 2^{n+1}}{(n+1)(n+2)}$$

Q6 Let $X = {}^{10}C_1^2 + 2({}^{10}C_2)^2 + 3({}^{10}C_3)^2 + \dots$, where ${}^{10}C_r, r \in \{1, 2, \dots, 10\}$ denote binomial coefficients. Then, the value of $\frac{1}{1430}X$ is

Q7 If $(1+x)^n = \sum_{r=0}^n {}^nC_r x^r$ then prove that

$$C_1^2 + 2 \cdot C_2^2 + 3 \cdot C_3^2 + \dots + n \cdot C_n^2 = \frac{(2n-1)!}{[(n-1)!]^2}$$

Q8 The value of ${}^{15}C_0^2 - {}^{15}C_1^2 + {}^{15}C_2^2 - \dots - {}^{15}C_{15}^2$ is
(A) 15 (B) -15
(C) 0 (D) 51

Q9 If $(1+x)^n = C_0 + C_1x + C_2x^2 + \dots + C_nx^n$,
then, $C_0^2 + C_1^2 + C_2^2 + C_3^2 + \dots + C_n^2$ is equal to

- (A) $\frac{n!}{n!n!}$
(B) $\frac{(2n)!}{n!n!}$
(C) $\frac{(2n)!}{n!}$
(D) None of these

Q10 $C_0C_r + C_1C_{r+1} + C_2C_{r+2} + \dots + C_{n-r}C_n =$
(A) $\frac{(2n)!}{(n-r)!(n+r)!}$



(B) $\frac{n!}{(-r)!(n+r)!}$

(C) $\frac{n!}{(n-r)!}$

(D) None of these

Q11 The value of r for which

$${}^{20}C_r \cdot {}^{20}C_0 + {}^{20}C_{r-1} \cdot {}^{20}C_1 + {}^{20}C_{r-2} \cdot {}^{20}C_2 + \dots + {}^{20}C_0$$

 ${}^{20}C_r$ is maximum is:

(A) 15

(B) 20

(C) 10

(D) 11

Q12 The value of $\sum_{r=0}^6 ({}^6C_r \cdot {}^6C_{6-r})$ is equal to

(A) 924

(B) 1024

(C) 1124

(D) 1324

Q13 The sum of $\sum_{i=0}^m ({}^{10}C_i) \cdot ({}^{20}C_{m-i})$ (where ${}^pC_q = 0$ if $p < q$) is maximum when m is:

(A) 5

(B) 10

(C) 15

(D) 20

Q14 If $(1+x)^n = c_0 + c_1x + c_2x^2 + \dots + c_nx^n$,then, $c_0c_1 + c_1c_2 + \dots + c_{n-1}c_n$ is equal to

(A) $\frac{(2n)!}{(n-1)!(n+1)!}$

(B) $\frac{(2n-1)!}{(n-1)!(n+1)!}$

(C) $\frac{(2n)!}{(n+2)!(n+1)!}$

(D) None of these

Q15 If ${}^{20}C_r$ is the co-efficient of x^r in the expansion of $(1+x)^{20}$, then the value of $\sum_{r=0}^{20} r^2 \cdot {}^{20}C_r$ is equal to

(A) 420×2^{19}

(B) 380×2^{19}

(C) 380×2^{18}

(D) 420×2^{18}

Q16 The value of $\binom{30}{0} \binom{30}{10} - \binom{30}{1} \binom{30}{11} + \binom{30}{2} \binom{30}{12} - \dots + \binom{30}{20} \binom{30}{30}$, where $\binom{n}{r} = {}^nC_r$ is

(A) $\binom{30}{10}$

(B) $\binom{30}{15}$

(C) $\binom{60}{30}$

(D) $\binom{31}{10}$



Answer Key

Q1 Hence Proved
Q2 Hence Proved
Q3 Hence Proved
Q4 Hence Proved
Q5 Hence Proved
Q6 646
Q7 Hence proved
Q8 (C)

Q9 (B)
Q10 (A)
Q11 (B)
Q12 (A)
Q13 (C)
Q14 (A)
Q15 (D)
Q16 (A)



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